

General Industries Hub

Enterprise Inventory Management

Auditor Manual

Read-only review across every site: reports, records, KPIs.

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Confidential - for authorized personnel only

GI Hub - Role booklet - Auditor Manual

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no
Two-stage approval	Store Keepers stage entries step-HOD ready and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal
Audit-first	Every action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.
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1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock	Permanent - stock figures move now
4	System	Sends Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then records the physical arrival from	Warehouse - Receiving
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals

Step	Role	Action	Lands in
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on	Permanent stock record; the

Three side-paths exist for when things do not go to plan: site delivery closes

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

Roles are ranked, but the ranking is not a straight line - two pairs sit side by side at the same level and differ only in what they are scoped to.

Level	Role	Scoped to
0	Store Keeper	One site
1	Warehouse User	One warehouse
1	Supervisor	One site
2	Head of Department	One site
3	Logistics	All sites
3	Auditor	All sites, read-only
4	Admin	Everything

A higher rank does not open a lower role's workspace. The Logistics Portal, Warehouse Portal, Entry Log, HOD Portal, Supervisor Portal, Material Estimator and Man-Hours pages are locked to their own role exactly - Logistics outranks a Head of Department but still cannot open the HOD Portal. Admin is the single exception and reaches every workspace deliberately, for support.

2.2 Page access matrix

Page	Store Keeper	Warehouse User	Supervisor	HOD	Logistics	Auditor	Admin
Live Dashboard	[X]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]
Entry Log	[OK]	[X]	[X]	[X]	[X]	[X]	[X]
HOD Portal	[X]	[X]	[X]	[OK]	[X]	[X]	[X] (hidden - uses Admin)
Logistics Portal	[X]	[X]	[X]	[X]	[OK]	[X]	[OK] (admin)
Warehouse Portal (NEW)	[X]	[OK]	[X]	[X]	[X]	[X]	[OK] (admin shadow, picks WH)
Admin	[X]	[X]	[X]	[X]	[X]	[X]	[OK]
Reports	[X]	[OK]	[OK]	[OK]	[OK]	[OK] (read + download only)	[OK]

Records (Inventory, ledgers, POs, PRs)	[X]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]
Supervisor Portal	[X]	[X]	[OK]	[X]	[X]	[X]	[OK] (admin
Material Estimator	[X]	[X]	[X]	[OK]	[X]	[X]	flow) [OK]
Man-Hours (NEW)	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]
Auditor view	[X]	[X]	[X]	[X]	[X]	[OK]	[OK]

> Exact-role locks: Entry Log, HOD Portal, Supervisor Portal, Logistics Portal, Warehouse Portal, Material Estimator and Man-Hours are locked to their own role - a higher-ranked role does not inherit access. Admin reaches them for support purposes.

2.3 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User (NEW)	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Reports, Live Dashboard, Burn Rate all site-locked.
HOD	Their own site only - but they can REQUEST material from
Logistics (NEW)	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Auditor (NEW)	All sites globally, read-only. Sits at level 3 so it is not site-locked - an auditor pinned to one site could not audit. It
Admin	All sites + all warehouses globally - has the "All Sites" filter on every multi-site view; warehouse picker in sidebar

when

shadowing the Warehouse Portal.

2.4 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD

Username	Password	Role
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics or warehouse_user accounts are seeded. Both roles are strictly admin-created - go to Admin Portal -> Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session

Element	Purpose	Notes
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
render_brand_header	Live Dashboard, Entry Log	Gold subtitle accent + current date
render_brand_header_hod	HOD Portal	Purple subtitle accent + current date
render_brand_header_admin	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

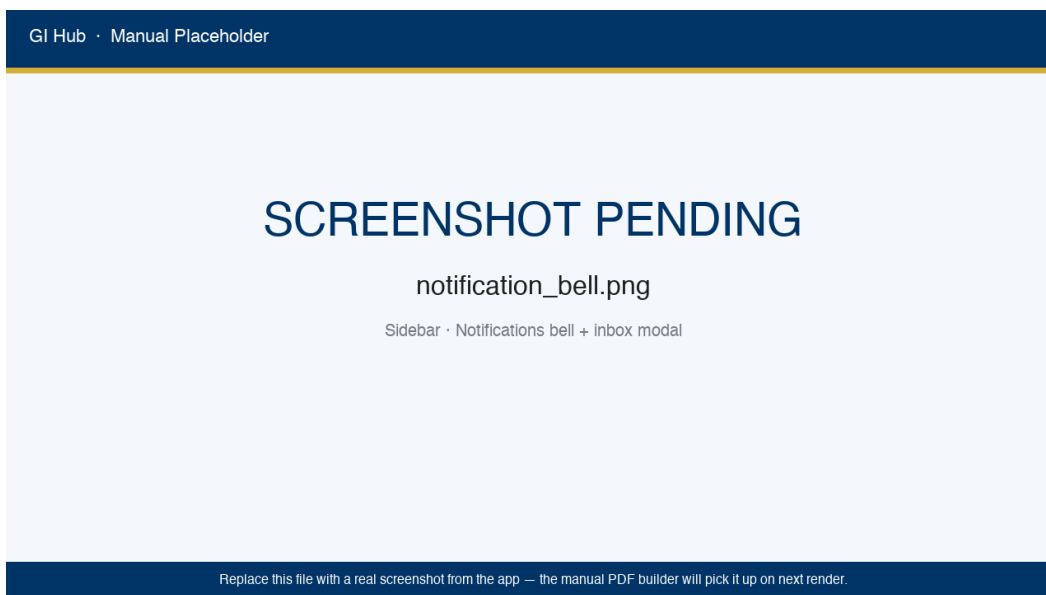
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

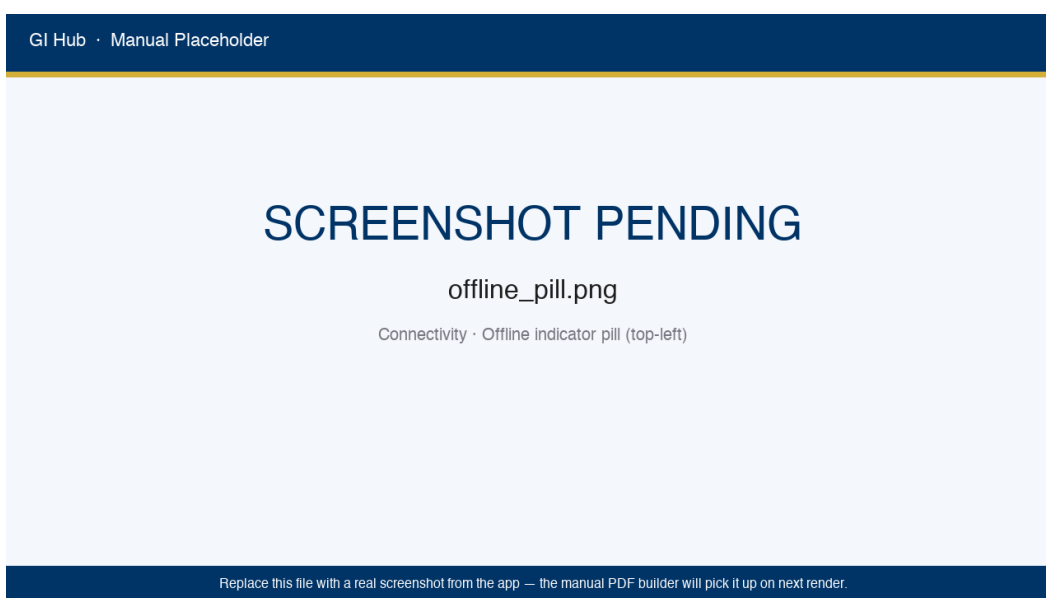
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope)

you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if a notification helper errors, the badge silently shows 0 instead of crashing the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by toggles in config.WHATSAPP_TRIGGERS (see §9).

3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

8. Reports Module - Detailed Reference

Available to: Supervisor, HOD, Admin. Site scope: locked for Supervisor + HOD; "All Sites" available to Admin.

8.1 Page header + 4 tabs

render_brand_header("Reports & Analytics"). Tabs:

1. Generate Report
2. Scheduled
3. AI Insights
4. Archive

8.2 Reports -> Generate Report

8.2.1 Date + filter row

Field	Purpose
From date	Start of date window
To date	End of date window
Site filter	Locked to own site for non-Admin; "All Sites" + dropdown for Admin
Format	PDF / Excel / CSV

8.2.2 Report type selector

A grid of 9 selectable cards:

Report	What it shows	Typical use
Daily Consumption	Every material issue in the window, grouped by work type	Reconciling a day's or week's usage against site activity
Monthly Summary	Per-item opening, issued, received and closing stock, with the SAR value of each	The month-end pack for finance
[!] Low Stock Alert	Materials below their minimum, with the shortfall	Building a reorder list
Burn Rate Analysis	30-day usage trend and days of supply	Deciding what to order before it runs out
Inventory Valuation	Stock value per item, highest first, with	Answering "what is on our shelves worth?"
Shelf-Life / Expiry	Items approaching expiry, critical and warning	The weekly walk-round to pull short-dated stock

Report	What it shows	Typical use
PR Status Report	Every purchase request with its stage and estimated cost	Chasing what has not arrived
[OK] FEFO Compliance	Whether issues took the oldest lot first,	Proving shelf-life discipline to an auditor
Full Audit Report	The complete activity history for the date range	Investigations, and formal audit evidence

8.2.3 > Generate Report button

Builds the report for the type, date window and site you chose. The page shows a loading shimmer while it works.

8.2.4 Preview section (after generation)

Element	Purpose
Report title + subtitle	The report type and site, above the date range covered
v Download button	Saves the report in the format you picked - PDF, Excel or CSV
Summary cards	The headline figures for the report, filled in automatically
Bar chart	Shown for the Burn Rate and Daily Consumption reports
Preview table	The rows themselves, with status colours

8.2.5 A note about apostrophes in downloaded files

Sometimes a downloaded Excel or CSV file shows an apostrophe in front of a value you typed - for example '=A1 where you typed =A1. Nothing is wrong, and nothing has been lost.

Spreadsheet programs treat anything starting with an equals sign, a plus, a minus or an at-sign as a calculation to run, not as words to display. If that text came from something a person typed into the system, opening the file could make the spreadsheet run instructions nobody intended. The apostrophe tells the spreadsheet "this is text, show it exactly as it is".

- It only ever appears in front of text that begins with one of those symbols.
- Numbers are never affected. Quantities, prices and totals stay numbers, and every total in the file still adds up correctly.
- Ordinary words, notes and material names are untouched.
- If you need the plain text back, delete the leading apostrophe in the cell.

You will see this most often on notes and remarks fields, since those are the places people type freely.

8.2.5 Email Delivery section

Field	Purpose
Recipients text input	Comma-separated emails
Send Email button	Generates and emails the report
WhatsApp button	Sends a brief summary via WhatsApp

8.3 Reports -> Scheduled

Manage automated recurring reports.

8.3.1 + New Schedule button (expands form)

Field	Purpose
Report Type	One of the 9
Frequency	Daily 06:00 / Daily 17:00 / Weekly Mon 07:00 / Monthly 1st
Format	PDF / Excel / CSV
Recipients	Comma-separated users

8.3.2 Schedule cards

Each schedule renders as a card with:

- Report icon + label
- Frequency - recipients
- Last run timestamp
- Active/Paused pill + toggle
- > Run Now button (manual trigger)
- Delete button

8.4 Reports -> AI Insights

If AI_ENABLED and Ollama is running with llama3.1:8b:

8.4.1 Header bar

+ "AI-Powered Inventory Analysis" + BETA pill + intro text. Shows "Analysing data..." shimmer during regen.

Button	Action
Regenerate	Re-runs all 5 fixed SQL probes + LLM commentary

8.4.2 5 Insight cards (collapsible)

Each card has a left-border in severity color, then:

Element	Purpose
Icon + Title	"Abnormal Consumption Spike - MAT-XXXX"
Severity pill (Critical/Warning/Positive)	Color-coded
Confidence % with progress bar	LLM-reported confidence
Right-side metric callout	Headline number with sub-label

Body paragraph	LLM-generated explanation grounded in the SQL probe result
Recommendations list	Up to 3 numbered actions
Share button	Shares with HOD team
[OK] Add to Actions button	Logs as a follow-up

The 5 fixed probes:

1. Abnormal consumption spike (vs trailing average)
2. Items approaching reorder
3. FEFO compliance rate
4. Procurement cost optimization (supplier consolidation)
5. Inventory health score

If Ollama is not running: a graceful fallback explains setup steps.

8.5 Reports -> Archive

Permanent record of previously generated reports.

8.5.1 Search row

Element	Purpose
Search archive text input	Substring match on name / type
Total counter caption	"N reports - X KB total"

8.5.2 Archive table

Columns: Report Name - Type pill (colored by report type) - Generated date - By - Format icon + label - Size - Actions

Per-row Actions:

- v Download
- Re-email
- Delete (audit-logged)

9. Automated Notifications - WhatsApp & Email & In-app Bell

The system automatically queues messages on key events. There are now THREE notification surfaces:

1. WhatsApp - messages queue and are sent by a background service, so a WhatsApp outage delays them but never loses them. There is a master on/off switch plus a per-event switch, so you can silence WhatsApp for one kind of event while leaving the in-app bell untouched for it.
2. In-app notifications bell (NEW in v3.0) - app_notifications table, surfaced via the sidebar bell described in §3.6. ALWAYS fires regardless of WhatsApp toggle.
3. Email - same Outlook / Mail.app / mailto: + SMTP paths as before.

9.1 WhatsApp triggers (auto-fire)

Event	Recipient	Trigger location	Notes
Store Keeper submits issue	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits receipt batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits stock	Site HOD	Entry Log Stock Count Submit	Includes variance + reason
Store Keeper performs FEFO override	Site HOD	Entry Log Add to Grid	Includes chosen vs FEFO-suggested lot + reason
HOD commits EOD	Site HOD (self)	post-commit	Only if low-stock items
HOD cross-site request created	All Admins	Submit cart	Includes target site + count
HOD cross-site request, > 5	Target Site HOD	Submit cart	Bulk escalation
Admin approves cross-site requests (batch)	Requesting HOD	Approve Selected	Includes approved items + admin notes
Admin approves cross-site requests (batch)	Target Site HOD	Approve Selected	"TRANSFER ORDER - pack
Admin rejects cross-site requests (batch)	Requesting HOD	Reject Selected	Includes rejection reason
Returnable item overdue	Borrower + Store Keeper	Background scheduler	Time-driven, not in current scope
New self-registration request	All Admins	Registration form	Includes username + role + site
Self-registration approved	Requesting user	Admin User Mgmt	"ACCESS GRANTED - Welcome"
Manual send	Free	HOD Notifications / Admin WhatsApp Console	Audit MANUAL_WHATSAPP

Event	Recipient	Trigger location	Notes
PR submitted to Logistics (v3.0)	Logistics role	HOD PR tab -> Submit PR(s) to Logistics	Event key pr_submitted_to_logistics
PO issued (v3.0)	Site HOD	Logistics -> Save PO	Event key po_issued
PO assigned to Warehouse	Warehouse users at that WH	Logistics -> Assign	Event key po_assigned_to_warehouse
Warehouse acknowledged (v3.0)	Logistics	Warehouse -> [OK] Acknowledge	Event key warehouse_acknowledged (off by default)
Warehouse received goods	Logistics	Warehouse -> Record receipt	Event key warehouse_received
Logistics approved (v3.0)	Site HOD	Logistics -> [OK] Approve DN	Event key dn_logistics_approved
DN HOD approved -> SK (v3.0)	Site SK	HOD -> DN Approvals -> [OK] Approve	Event key dn_auto_generated (reuses dn_received_by_sk)
DN received at site (v3.0)	Logistics + Warehouse	SK -> [OK] Mark as Received	Event key dn_received_by_sk (off by default)
Reschedule requested (v3.0)	Logistics	Warehouse / HOD -> Request reschedule	Event key reschedule_requested
Reschedule decided (v3.0)	Requester	Logistics -> [OK] Approve / [X] Reject	Event key reschedule_decided
Vendor return raised (v3.0)	Logistics	Any role -> <- Raise return	Event key vendor_return_raised
PR force-closed (v3.0)	Admin + originating Site HOD	Logistics -> Force-Close	Event key pr_force_closed, severity critical
PO force-closed (v3.0)	Admin + originating Site HOD	Logistics -> Force-Close	Event key po_force_closed, severity critical
Delivery reminder T-2 / T-1 / T-0 (v3.0)	Logistics + HOD + Warehouse (per DN)	Daily sweep job - see §9.3	Event key delivery_reminder_t_minus_2_minus_1_zero

9.2 Email triggers

Event	Trigger location	Mechanism
Logistics PR follow-up	HOD PR tab -> Draft Outlook Email	Mac: AppleScript-driven Mail.app with HTML table. Windows: COM Outlook with HTMLBody. Fallback: mailto:
EOD report email	mailer.py SMTP	Configurable via .env (SMTP_SERVER, SMTP_USER, SMTP_PASS)
Report delivery (scheduled)	Reports -> Scheduled	Same SMTP pipeline

9.3 In-app notifications bell (NEW in v3.0)

See §3.6 for the UI. Backed by the `app_notifications` table; queried with role + site + warehouse scoping. Per-event severity (info / warning / success / critical) drives the colour-coded left border on each card.

9.4 Delivery reminder daily sweep (NEW in v3.0)

Once a day the system looks ahead at upcoming deliveries and sends reminders two days before, one day before, and on the day itself:

Watched	When	Severity
<code>purchase_orders.Expected_Delivery</code>	T-2 / T-1 / T-0	warning / warning / critical
<code>delivery_notes.DN_Date</code>	T-2 / T-1 / T-0	warning / warning / critical

PO reminders ping: Logistics + originating Site HOD. DN reminders ping: Logistics + Site HOD + Warehouse user(s) at the receiving warehouse.

Idempotency - the sweep cannot double-fire for the same target on the same day. Two guards:

1. `UNIQUE(ref_type, ref_number, target_date, offset_days)` on the `delivery_reminders_sent` table.
2. Day-marker stored in `app_settings.delivery_reminders_last_run` - the 60-sec worker poll loop skips the sweep query entirely on second-and-later ticks of the same day.

Restarting the worker mid-day is safe. Re-running the sweep manually on the same day fires zero new notifications.

Customising the cadence - the two-day, one-day and same-day pattern is currently fixed. Making it configurable per site is on the backlog; ask if you need a different rhythm.

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (PK), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override, Work_Type	- subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current_Stock} = \text{Total_Received} - \text{Total_Consumed} - \text{Total_Returned} - \text{computed live in v_site_stock view.}$

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	API tokens for offline PWA
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned /
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	read_at IS NULL (unread) -> timestamp (read)
delivery_reminders_sent	T-2/T-1/T-0 idempotency log	(terminal - UNIQUE constraint)

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	Substring RL-, RUBBER LINING, RUBBER-LINING in Material_Code OR Description
BL	Brick Lining	Substring BL-, BRICK LINING, BRICK-LINING, BRICK MATERIAL
NULL	Neither family	Default

Logic in the system. NEVER a combo string - RL takes precedence if both tokens are present.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> Access Control -> Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13

16. Cross-Role Procurement Walk-through

A single happy-path narrative threading all five roles together, useful for onboarding.

16.1 The scenario

Site GI-PS01 needs 50 RL panels and 20 BL bricks. Total 3 working days from PR raise to physical receipt.

16.2 Hour by hour

When	Who	What
Day 1 - 8:00 AM	Site HOD	Opens HOD Portal -> Purchase Requests. Creates 2 PR lines (50 RL panels, 20 BL bricks) with PR Number 3000099999, WBS Number 4003951, Network 4003951-PROJ-A, Plant
Day 1 - 8:15 AM	Site HOD	Opens the Delivery PRs Day 1 Logistics Portal expander -> multi-selects PR 3000099999 -> Submit Selected to Logistics.
Day 1 - 8:15 AM	(auto)	pr_submitted_to_logistics notification -> Logistics inbox (red badge). WhatsApp ping if enabled.
Day 1 - 9:00 AM	Logistics	Sees PR 3000099999 in Incoming PRs queue. Drills in - 2 lines, RL + BL.
Day 1 - 9:30 AM	Logistics	Switches to Create PO -> manual entry. Picks vendor "Carborundum Universal" (0000110341), Inco/Payment auto-fill, fills PO Number 4720033030, Unit_Price per line,
Day 1 - 9:30 AM	(auto)	PO auto Price computed. Save PO. logistics_status='in_po' -> disappear from Logistics queue. Site HOD gets po_issued notification
Day 1 - 10:00 AM	Logistics	Assign to Warehouse -> picks PO 4720033030 + Warehouse WH-A + Expected Delivery Day 2 evening. Assign.

Day 1 - 10:00 AM	(auto)	po_assigned_to_warehouse notification -> WH-A users' inbox. WhatsApp ping if enabled.
Day 1 - 11:00 AM	Warehouse user (WH-A)	Incoming Assignments -> [OK] Acknowledge assignment #N. Goes about their day.
Day 2 - 5:00 PM	Warehouse user	Physical truck arrives from vendor. Receive Goods -> assignment #N -> type 50 in Receive Now for the RL line, 20 for the BL line. Record receipt.
Day 2 - 5:30 PM	Warehouse user	Prepare DN -> pick PO 4720033030, destination GI-PS01. Types Ship Qty 50 for RL line, attempts to add BL line - rejected (RL/BL strict separation). Saves DN-WH-A-20260617-001 for
Day 2 - 5:35 PM	Warehouse user	Repeats: new DN with only the BL line.
Day 2 - 5:40 PM	Warehouse user	Outbound DNs -> submits both drafts to Logistics.
Day 2 - 5:45 PM	Logistics	Sees DN-WH-A-20260617-001 + 002 in their DN approval queue. Confirms delivery date Day 3 AM. Approves both.
Day 2 - 5:45 PM	(auto)	DN status flips to pending_hod. Site HOD gets dn_logistics_approved notification.
Day 2 - 6:00 PM	Site HOD	Logs in, sees red bell badge 2 unread. Opens inbox -> drills into the DNs via
Day 2 - 6:00 PM	(auto)	DN status pending_sk. Two mirror rows in pending_receipts.status='pending_sk' . Site SK gets dn_auto_generated notification.
Day 3 - 8:30 AM	Truck arrives at site GI-PS01	
Day 3 - 8:45 AM	Site SK	Opens Entry Log -> Receipt Staging. The new Incoming Delivery Notes from Warehouse expander is open at the top, showing both DNs. Inspects both, confirms physical match, clicks [OK] Mark as Received on each.

Day 3 - 8:45 AM	(auto)	Two receipts rows written (with DN_Number, Warehouse_ID, PO_Number_Source). Both DNs flip to received. pending_receipts mirror rows deleted. Inventory cache busted. Live Dashboard shows the new stock immediately.
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16.3 Side-path examples

A site rejects a DN

Same scenario, but on Day 2 at 6:00 PM, Site HOD inspects the line preview on DN-002 and notices the BL qty doesn't match the PR. Rejects DN-002 with note "BL qty in DN is 25, PR asked for 20 - bounce back".

- DN-002 status -> rejected, rejection_reason stored
- Warehouse user gets the rejection notification with the reason
- The BL line on the PO frees up qty (the over-ship guard recalculates available since DN-002 is no longer "live")
- Warehouse prepares DN-003 with correct qty

Logistics force-closes the PR mid-flight

Project cancelled by management on Day 1 evening:

- Logistics -> Force-Close -> radio = PR -> reason "Project cancelled per CFO email 2026-06-16" -> Force-close PR
- Admin + Site HOD see critical-severity notification in their bell
- The pending PO can either stay (if vendor already shipped) or also be force-closed
- Site HOD's In-Transit -> Force-closures sub-tab shows the closure with the reason

Warehouse asks for reschedule

Receiving day Truck doesn't show up - vendor delay:

- Warehouse -> Outbound DNs -> drill into DN-001 -> Request reschedule -> new date Day 4 -> reason "Vendor delivery delayed 24h confirmed by vendor email"
- Logistics -> Reschedules -> reviews -> [OK] Approve with notes "Confirmed with vendor"
- PO Expected_Delivery, po_assignments.Expected_Delivery, AND DN_Date all flip to Day 4 in one transaction
- Site HOD sees the decision notification

20. Auditor (View-Only) Manual

Added 2026-08-03.

20.1 What the Auditor role is for

The Auditor is for anyone who must see the numbers without being able to move them - an internal auditor, a client-side QA reviewer, a finance reviewer, a new manager still learning the system, or a consultant given temporary access.

It is a real role like any other, with one difference that is absolute:

> An Auditor can read everything its level reaches and can change nothing, > anywhere, ever.

That is not a UI convention. It is enforced on the server for every request.

20.2 What an Auditor can see

Area	What it shows
Dashboard	Every KPI, chart and category breakdown, across all sites
Stock	Current stock by site, minimum levels, expiry
Records	Inventory, Receipts, Consumption, Returns, Lots, Purchase
Reports	Every report, as Excel, PDF or CSV
Lining Coverage	The read-only coverage analysis
Documents	The SOP and User Manual PDFs, QR/label sheets
Account	Own security settings (password-manager 2FA, phone number)

An Auditor is not site-locked. Every other role below Logistics sees only its own site; an auditor sees all of them, because an auditor confined to one site cannot audit the business.

20.3 What an Auditor cannot do

Everything that changes data. The Data Entry forms, Master Data editors, Bulk Excel Import, HOD Approvals, the Logistics and Warehouse portals, the Supervisor portal and the Admin console do not appear in the sidebar at all - they are not hidden buttons on a visible page, the pages are simply not reachable.

On pages an Auditor can open, actions that would change something are disabled rather than hidden, with a tooltip explaining why:

> Your account is view-only (Auditor) - this action changes data.

On the Reports page, for example, Excel / PDF / CSV downloads work normally (they only read), while Archive, WhatsApp, New schedule, Run now and Delete are greyed out.

A "View only" tag sits permanently in the header beside the user name, so there is never any doubt about which kind of account is signed in.

20.4 What an Auditor CAN still do

Because they are read-only, not crippled:

- Download every report in all three formats.
- Use the CmdK / Ctrl-K command palette to jump to any page or material.
- Ask the Hub Assistant questions (it answers from the Auditor's own manual chapters - see §20.6).
- Manage their own account: enable or disable two-factor authentication and change their own phone number. These touch only their own user record.

20.5 The security model (for Admins)

Worth understanding before you hand the role out.

The restriction is enforced once, centrally, rather than screen by screen. Any request that would change data from a view-only account is refused before it reaches the feature at all, and the user is told plainly that their account is view-only and the action is not permitted.

Why that design matters to you: a per-endpoint permission check is only as reliable as the developer who remembers to add it, and the one that gets forgotten fails open - the write succeeds and nothing looks wrong. Keying on the method inverts that, so a feature added next year is closed to Auditors from the moment it is written. 126 of the system's 143 state-changing endpoints are blocked; the 17 that are not are sign-in, sign-out, token refresh, the user's own 2FA and phone-change flows, and read-only calculation endpoints that use POST only because the request is too large for a URL.

Two consequences to be aware of:

- An Auditor probing a URL that does not exist gets 403, not 404. The guard runs before routing, so it cannot be used to map the system.
- Blocking is by role, not by page. Even if an Auditor somehow reached a write page, every button on it would still fail server-side.

20.6 Creating an Auditor account

Admin Portal -> Users -> Add user, and pick Auditor (view-only) in the Role dropdown. Leave the Site blank - Auditors are global by design, and the form does not ask for one.

Users can also request the role themselves on the registration screen; it still needs Admin approval like any other request.

20.7 The Hub Assistant for Auditors

The assistant answers from chapters 1, 2, 3, 8, 9, 10, 11, 12, 16 and 20 - orientation, the reports module, notifications, the data model, the glossary, the FAQ and the cross-role procurement walk-through. That is

deliberately the set matching the pages an Auditor can open.

Ask it about a report, a status code or what a field means and it will answer. Ask it how to post a consumption entry and it will tell you that is outside the read-only Auditor view - because the operational chapters for roles that can post entries are never loaded into an Auditor's context in the first place.

21. 2026-08 Feature Update - What Changed

Everything in this chapter is live. Grouped by what you would notice.

21.1 Reports and exports look the same everywhere now

Every Excel export - not just the Material Estimator's - now carries the branded layout: the GI logo, a "Report Generated / Generated By" band, a title bar and a framed, filterable table with the header frozen. Previously only three SME workbooks looked like that and everything else came out as a bare grid.

> If you have automation reading these files: the header row is now row 6 > and data begins on row 7 (rows 1-4 are the logo and meta band, row 5 the > title bar). Anything that assumed a header on row 1 needs adjusting.

PDF reports no longer overlap or cut off text. Columns are now measured against their actual content and share the page fairly, and long values wrap onto more lines instead of being truncated. Previously every column got an equal slice of the page regardless of need, so a long material description was drawn on top of the next column while Date and UOM wasted most of theirs - and anything past 24 characters was silently deleted.

QR badge and bin labels were fixed the same way: a long employee name used to run past the edge of its sticker onto the one beside it.

21.2 The Material Estimator understands your order data correctly

Ordered_Qty is the TOTAL quantity procured for the project, and Available_Qty is the part of it that has already arrived - available is a subset of ordered, not a separate pile beside it.

The estimator used to add the two together, which double-counted every unit already on the shelf. On real data that understated the buy list by 22,951 units across 22 of 30 materials, and on GI-8005763 - where all 143,000 ordered units had arrived - it read 286,000 and reported nothing to buy against a demand of 152,685, hiding a 9,685-unit shortage completely.

What you will see in the UI:

- Second-tier quantities now read "Pending Delivery" - the part of the order still on its way - never just "Ordered".
- Second-tier coverage reads "When delivered".
- Reports carry a Total Procured column beside them.

Feasibility is unchanged and still judges physical stock only: a tank cannot be built with a purchase order.

21.3 A view-only Auditor role

See §20.

21.4 You are signed out after 30 minutes of inactivity

Leave the app untouched for 30 minutes and it signs you out and returns you to the login screen. A warning appears two minutes before with a button to stay signed in, and any real activity - a click, a keystroke, a scroll - resets the clock silently.

This matters most on shared site terminals, where an unattended session was previously good until the browser was closed.

21.5 Search anything with CmdK

Press CmdK (Mac) or Ctrl-K (Windows) anywhere in the app.

The palette already jumped to pages; it now also searches your inventory live. Type a SAP code, part of a material name or a material code and the matching items appear under a Materials heading - pick one and you land on its Material Intelligence card. Results are scoped to what your role may see.

21.6 The Hub Assistant is faster and more accurate

The assistant used to be handed its entire allowed portion of this manual on every single question. For an Admin that was the whole ~180 KB document, which is slow to process; for everyone else it was the first 800 characters of each chapter, which is the wrong 800 whenever the answer is further down.

It now finds the passages that actually match your question and reads only those - a 97.7 % smaller prompt for Admins, and noticeably better answers for everyone, because the relevant paragraph is now present in full instead of being cut off.

A parsing bug was fixed at the same time: shell comments inside the code blocks in the Operations chapter (# 1. Pull the new code) were being read as chapter headings, which silently replaced chapters 1-4 - Introduction, Roles & Permissions, Login, and the Store Keeper Manual - with fragments of a restore script. Ask "how do I log in?" today and you get §3.1, the login screen.

The role filter is unchanged and still applied before retrieval: a Store Keeper's context physically cannot contain an Admin chapter.

21.7 Serialised tool and equipment tracking

Two hammers share one item code, so scanning either label used to resolve to the same inventory line - the system could not tell you which one was in your hand or where the other had gone. Assets solves that.

Each physical item gets its own record, identified by its serial number. A scan now resolves three ways, and the order is deliberate:

What the label carries	What you get
A serial number or asset tag	That exact item, immediately - there is nothing to choose
Only the item code, and several units exist	A "which one is in your hand?" list to pick from

What the label carries	What you get
Only the item code, and no units are registered	The ordinary material view - the system does not invent an

Recording where something is. Open the item, tap **Move**, and either pick a rack or describe the place in words - not everything lives on a shelf, and "loaded on truck 4771" is a real answer. If the device can supply coordinates, they are captured alongside the update, never in front of it: a declined permission or a yard with no signal still records the move, with the coordinates simply absent. A recorded position becomes a map link on the item.

Status covers both what condition a tool is in - working, not in use, under repair - and where it sits in the stores workflow. The full history of every move is kept and is never overwritten, so "where has this been" is a question with an answer.

> Coordinate capture needs a secure connection. Over plain HTTP the browser > refuses to supply a position - the move still saves, without coordinates. If > your team is on the installed app or the company web address, this is already > the case.

> Your spreadsheet seeds this; the app owns it. A location typed in the > Consumption Log creates the asset on the next sync. From then on, what somebody > recorded in the app wins: a re-sync will not move a tool, revert its status, or > erase a captured position.

21.8 The warehouse rack locator

"Which rack is this in?" now has an answer. Search by material name, by code, or by scanning a label, and the system tells you the shelf to walk to - ordered so that the first result is where to go first, even when a material is stocked in three places.

- A material you stock but nobody has located comes back marked not located, rather than being left out. Silence would read as "we do not stock it", which sends somebody out to buy what is already on a shelf.
- Scanning the rack itself answers the reverse question - everything that is meant to be on that shelf. That turns a stock count from a hunt into a checklist.
- The locator is built into the CmdK search, so it is two keystrokes from anywhere in the application.

21.9 The Material Estimator handles messy tank references

The Consumption Log records a tank reference typed by hand, and the same vessel appears written four different ways. The system now recognises those as one vessel automatically - that alone accounted for 58 of 103 rows.

Where a reference is genuinely ambiguous, it refuses to guess. TNK-091 exists on both TRAIN J and TRAIN K; either answer would look completely plausible in every report afterwards, so those rows wait in a queue for someone who knows. Resolve the reference once on the Actual Consumption tab and every row waiting on it is tagged.

21.10 For administrators - local tooling

Two maintenance scripts ship with the system, documented in the administrator runbook: one puts the database and the public connection to sleep and wakes them again for a laptop that is not always on, and one takes and restores database snapshots, with an option to run nightly at 02:00.

> The previous nightly backup had been failing silently for 25 consecutive > nights - it pointed at a script that a repository reorganisation had moved. > Anyone relying on it had no local backups at all. The replacement is verified > by actually restoring into a scratch database on every change, which is the > only check that means anything.

21.11 Downloaded files now protect you from your own spreadsheet

If you type something that begins with an equals sign, a plus, a minus or an at-sign into a text field - a remark, a note, a reason - you may notice a small apostrophe in front of it when you open the downloaded Excel or CSV file.

That is deliberate, and nothing has been lost.

Spreadsheet programs treat a value beginning with one of those symbols as a calculation to run, not as words to show. That is fine when you write the formula yourself. It is not fine when the text arrived from somewhere else: a note typed at one site, exported by a manager and opened on their computer, would be run by their spreadsheet without anyone choosing to. The apostrophe tells the spreadsheet to show the text exactly as it was written instead.

What this means in practice:

- Numbers are never touched. Quantities, prices, areas and every total in every report stay real numbers, and every total still adds up.
- A plain number that happens to be negative, such as -5, is left exactly as it is - it is a number, not a calculation.
- Ordinary text - material names, supplier names, ordinary notes - is untouched.
- If you want the plain text back in your own copy, delete the apostrophe in that cell.

You are most likely to see this on remarks and reason fields, because those are the places people type freely. It applies to every download the system produces: reports, master-data exports and the Material Estimator workbooks alike.

Document end

This manual covers every page, tab, button, table and field in the General Industries Hub: the site workspaces (sections 4 to 7), the Reports module (section 8), the Logistics and Warehouse portals and the procurement chain (sections 14 to 16), hosting and operations (section 17), the Material Estimator (section 18), Man-Hours and Labor Tracking (section 19), the Auditor role (section 20), and the asset-tracking and warehouse-locator features added in 2026-08 (section 21).

For day-to-day operating procedure across all roles, see the Standard Operating Procedure document, which is shorter and organised by task rather than by screen.

Printed and PDF copies

Every role has its own booklet containing only the chapters that apply to them - a Store Keeper's booklet does not include the Logistics portal, and is a fraction of the length. Admin Portal -> Settings offers each of them as a download, and the full manual alongside.

If you are handing the system over to a new team, print the role booklets rather than this complete document. People read twenty relevant pages and skip two hundred.